Nowcasting

Workbook

Workshop 5– Forecast Combination, Evaluation, and Realistic Forecasts in EViews: India Case Study



IMF Institute for Capacity Development

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Section 0: EViews Command Cheat Sheet

You may refer to the text file ***[cheatsheet] w-5.rtf*** for a "cheat sheet" of EViews commands used in this workshop (you may copy and paste the specific command lines instead of typing them).

You may open the text file using EViews by clicking "File => Open => Text File".



```
' W-5 Forecast Evaluation and Combination
' ========================================

' Section I: Nowcast Combinations
'--------------------------------
' simple average

' To calculate the nowcast combination (i.e., 2023Q2):

smpl @all
genr nowcast_rgdp_fc = (nowcast_rgdp_bridge + nowcast_rgdp_3prf +
nowcast_rgdp_midas + nowcast_rgdp_midas_pc + nowcast_rgdp_umidas
+nowcast_rgdp_umidas_pc +nowcast_rgdp_elastic+nowcast_rgdp_lasso
+nowcast_rgdp_ridge+nowcast_rgdp_mf_var)/10.0

' To calculate the in-sample forecast combination (i.e., 2018Q1-2023Q1):

smpl 2018Q1 2023Q1

smpl 2018Q1 2023Q1
genr rgdp_f_fc = (rgdp_f_bridge + rgdp_f_midas + rgdp_f_umidas +
rgdp_f_midas_pc + rgdp_f_umidas_pc +rgdp_f_3prf + rgdp_f_mf_var+
rgdp_f_ridge+ rgdp_f_lasso+ rgdp_f_elastic)/10.0

' In-Sample Forecast Evaluation

smpl 2018Q1 2023Q1

rgdp.forceval rgdp_f_fc rgdp_f_bridge rgdp_f_midas rgdp_f_umidas
rgdp_f_midas_pc rgdp_f_umidas_pc rgdp_f_3prf rgdp_f_mf_var rgdp_f_ridge
rgdp_f_lasso rgdp_f_elastic

' Trimmed Mean
' to calculate the nowcast combination

smpl @all
genr nowcast_rgdp_fc_trim = (nowcast_rgdp_bridge + nowcast_rgdp_umidas +
nowcast_rgdp_midas_pc + nowcast_rgdp_umidas_pc +
nowcast_rgdp_3prf+nowcast_rgdp_3prf + nowcast_rgdp_ridge+
nowcast_rgdp_lasso+ nowcast_rgdp_elastic)/8.0

' to calculate the in-sample forecast combination

smpl 2018Q1 2023Q1

genr rgdp_f_fc_trim = (rgdp_f_bridge + rgdp_f_umidas + rgdp_f_midas_pc +
rgdp_f_umidas_pc +rgdp_f_3prf + rgdp_f_ridge+ rgdp_f_lasso+
rgdp_f_elastic)/8.0

' In-Sample Forecast Evaluation

smpl 2018Q1 2023Q1

rgdp.forceval rgdp_f_fc rgdp_f_fc_trim rgdp_f_bridge rgdp_f_midas
rgdp_f_umidas rgdp_f_midas_pc rgdp_f_umidas_pc rgdp_f_3prf rgdp_f_mf_var
rgdp_f_ridge rgdp_f_lasso rgdp_f_elastic
```

Section I: Nowcast Combinations and in-Sample Forecast Evaluation

Averaging across separate forecasts using a user-specified weighting scheme is called a "combined" forecast. Combined forecasts have become popular recently because they often have better forecasting performance than the separate forecasts that go into the average.

We can calculate combination forecasts using the nowcasts generated in the previous sections. Recall that we have 11 competing nowcasts for real GDP for 2023Q2, namely:

1. Bridge
2. MIDAS
3. U-MIDAS
4. MIDAS with PC
5. UMIDAS with PC
6. 3PRF
7. MF-VAR
8. Ridge
9. Lasso
10. Elastic Net

While there exist several weighting schemes to average these, this workshop will use simple weights (i.e., 1/10 for each series) to obtain a combined forecast (or nowcast) and trimmed weights.

We will continue with "**nwc.wf1**" for this workshop.

First create a pagefile called, **"SC"**, that gathers all relevant estimates for comparison.

```
pageselect ML
pagecopy(page=SC)
```

Next, in the **"SC"** pagefile, and issue the following commands in the command line area:

To calculate the nowcast combination (i.e., 2023Q2):

```
smpl @all

genr nowcast_rgdp_fc = (nowcast_rgdp_bridge + nowcast_rgdp_3prf +
nowcast_rgdp_midas + nowcast_rgdp_midas_pc + nowcast_rgdp_umidas
+nowcast_rgdp_umidas_pc +nowcast_rgdp_elastic+nowcast_rgdp_lasso
+nowcast_rgdp_ridge+nowcast_rgdp_mf_var)/10.0
```

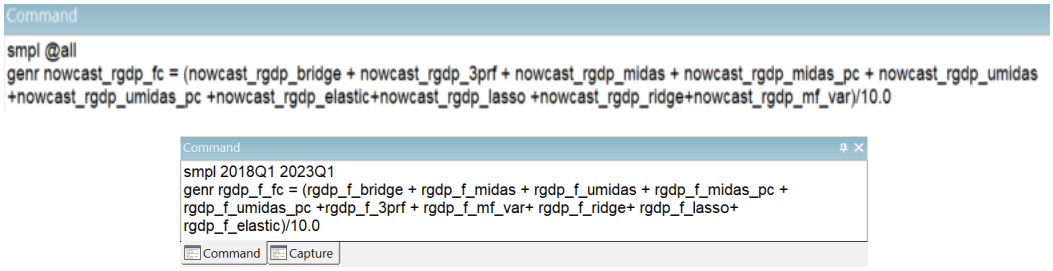
To calculate the in-sample forecast combination (i.e., 2018Q1~2023Q1):[1]

```
smpl 2018Q1 2023Q1

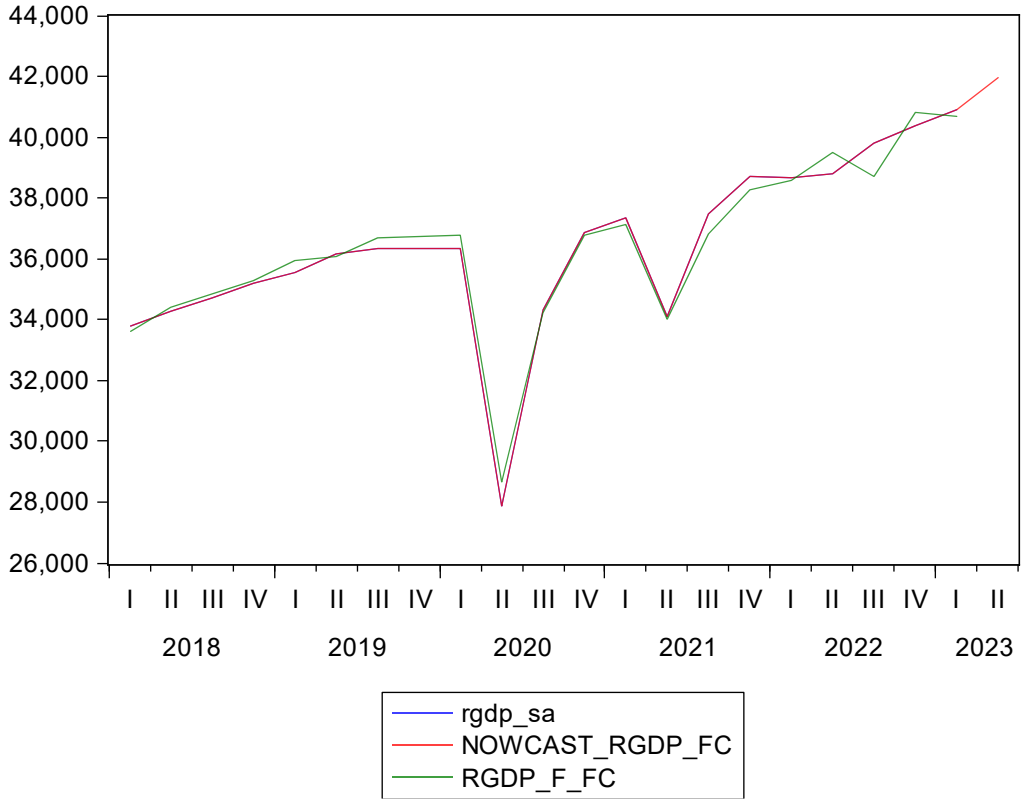
genr rgdp_f_fc = (rgdp_f_bridge + rgdp_f_midas + rgdp_f_umidas + rgdp_f_midas_pc +
rgdp_f_umidas_pc +rgdp_f_3prf + rgdp_f_mf_var+ rgdp_f_ridge+ rgdp_f_lasso+
rgdp_f_elastic)/10.0
```

This command averages the eleven (or ten) nowcasts using simple averaging and saves the result in the **nowcast_rgdp_fc** and **rgdp_f_fc** series in the **Quarterly** pagefile (Figure 1)

Figure 1: Combination Forecasts/Nowcasts for Real GDP



[1] Note, we will exclude in-sample forecast for DFM. We will evaluate its performance using realistic forecast evaluation in the next section.



We can compare the in-sample forecasting performance of the combined forecast to each forecast that makes up the average by using the **forceval** function, viz:

smpl 2018Q1 2023Q1

rgdp.forceval rgdp_f_fc rgdp_f_bridge rgdp_f_midas rgdp_f_umidas rgdp_f_midas_pc rgdp_f_umidas_pc rgdp_f_3prf rgdp_f_mf_var rgdp_f_ridge rgdp_f_lasso rgdp_f_elastic

The **forceeval** function compares (actual) real GDP to the in-sample forecasts generated for 2018Q1-2023Q1 (Figure 2).

Figure 2: Nowcast/Forecasting Performance, 2018Q1-2023Q1

Evaluation statistics

Forecast	RMSE	MAE	MAPE	SMAPE	Theil U1	Theil U2
RGDP_F_FC	438.7089	337.7370	0.931804	0.930881	0.006011	0.146833
RGDP_F_BRIDGE	390.1466	284.0905	0.775660	0.775343	0.005345	0.129826
RGDP_F_MIDAS	378.9160	283.9129	0.768578	0.768833	0.005191	0.123899
RGDP_F_UMIDAS	531.7762	425.9771	1.149257	1.150015	0.007293	0.177861
RGDP_F_MIDAS_PC	563.8586	447.4524	1.205925	1.206028	0.007731	0.187706
RGDP_F_UMIDAS_PC	563.8586	447.4524	1.205925	1.206028	0.007731	0.187706
RGDP_F_3PRF	462.9435	326.7615	0.890322	0.889380	0.006341	0.152811
RGDP_F_MF_VAR	1086.498	635.3160	1.871404	1.832580	0.014856	0.376034
RGDP_F_RIDGE	414.0996	299.8261	0.822532	0.822333	0.005674	0.138270
RGDP_F_LASSO	396.5671	267.4746	0.733957	0.732986	0.005434	0.132163
RGDP_F_ELASTIC	396.6165	267.4951	0.734010	0.733039	0.005434	0.132181

Lastly, researchers often recommend using a "trimmed mean" instead of the simple average to calculate a combination forecast. The trimmed mean is the average forecast excluding the series that have the best and worst forecasting performances. The aim is to reduce the influence of outliers on the average nowcast.

For the example shown above, based on the RMSE, we would remove **rgdp_f_mf_var** and **rgdp_f_midas** from the calculation of the average nowcast since these nowcasts have, respectively, the worst and best forecasting performance.

The trimmed mean would be calculated as follows:

To calculate the nowcast combination (i.e., 2023Q2):

smpl @all

genr nowcast_rgdp_fc_trim = (nowcast_rgdp_bridge + nowcast_rgdp_umidas + nowcast_rgdp_midas_pc + nowcast_rgdp_umidas_pc + nowcast_rgdp_3prf + nowcast_rgdp_ridge+ nowcast_rgdp_lasso+ nowcast_rgdp_elastic)/8.0

To calculate the in-sample forecast (i.e., 2018Q1~2023Q1):

smpl 2018Q1 2023Q1

genr rgdp_f_fc_trim = (rgdp_f_bridge + rgdp_f_umidas + rgdp_f_midas_pc + rgdp_f_umidas_pc +rgdp_f_3prf + rgdp_f_ridge+ rgdp_f_lasso+ rgdp_f_elastic)/8.0

Again, we can compare the in-sample forecasting performance of the combined forecast (both simple mean and trimmed mean) to each forecast that makes up the average by using the **forceval** function, viz:

rgdp.forceval rgdp_f_fc rgdp_f_fc_trim rgdp_f_bridge rgdp_f_midas rgdp_f_umidas rgdp_f_midas_pc rgdp_f_umidas_pc rgdp_f_3prf rgdp_f_mf_var rgdp_f_ridge rgdp_f_lasso rgdp_f_elastic

The in-sample forecasting performance of the trimmed mean is shown in Figure 3.

Figure 3: Nowcast/Forecasting Performance, 2018Q1-2023Q1 (Trimmed Mean Included)

Forecast Evaluation
Date: 02/06/25 Time: 16:19
Sample: 2018Q1 2023Q1
Included observations: 21
Evaluation sample: 2018Q1 2023Q1
Number of forecasts: 12

Evaluation statistics

Forecast	RMSE	MAE	MAPE	SMAPE	Theil U1	Theil U2
RGDP_F_FC	438.7089	337.7370	0.931804	0.930881	0.006011	0.146833
RGDP_F_FC_TRIM	418.9754	326.9347	0.887219	0.887519	0.005742	0.138642
RGDP_F_BRIDGE	390.1466	284.0905	0.775660	0.775343	0.005345	0.129826
RGDP_F_MIDAS	378.9160	283.9129	0.768578	0.768833	0.005191	0.123899
RGDP_F_UMIDAS	531.7762	425.9771	1.149257	1.150015	0.007293	0.177861
RGDP_F_MIDAS_PC	563.8586	447.4524	1.205925	1.206028	0.007731	0.187706
RGDP_F_UMIDAS_PC	563.8586	447.4524	1.205925	1.206028	0.007731	0.187706
RGDP_F_3PRF	462.9435	326.7615	0.890322	0.889380	0.006341	0.152811
RGDP_F_MF_VAR	1086.498	635.3160	1.871404	1.832580	0.014856	0.376034
RGDP_F_RIDGE	414.0996	299.8261	0.822532	0.822333	0.005674	0.138270
RGDP_F_LASSO	396.5671	267.4746	0.733957	0.732986	0.005434	0.132163
RGDP_F_ELASTIC	396.6165	267.4951	0.734010	0.733039	0.005434	0.132181

This latest evaluation table shows that the trimmed mean does not improve the in-sample performance compared to using the simple mean. MIDAS model still outperforms the trimmed mean combination forecast. Figure 4 compares the nowcast estimates of all models.

Figure 4: Nowcast Comparison, 2023Q2 (in YoY growth rate %)

Models	2023Q2
3PRF	8.48
BRIDGE	7.99
DFM	5.17
ELASTIC	7.97
LASSO	7.97
MF_VAR	7.78
MIDAS	8.39
MIDAS_PC	8.56
RIDGE	8.05
UMIDAS	8.65
UMIDAS_PC	8.56
FC	8.22
FC_TRIM	8.28

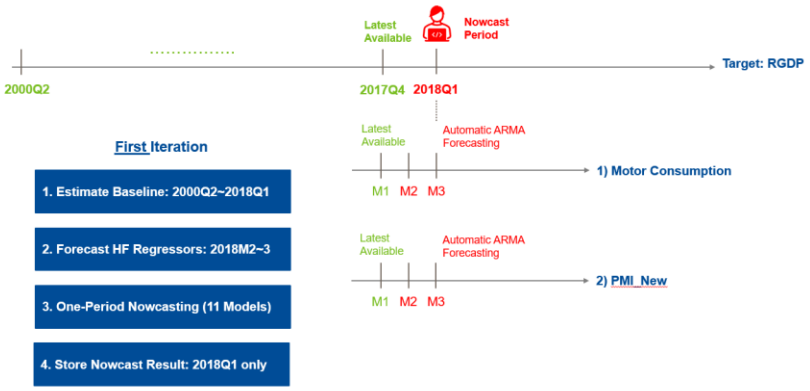
The best in-sample performing model (i.e., MIDAS) predicts a 2023Q2 y-on-y growth rate of 8.39 percent. The in-sample performance measures the model's goodness of fit. However, out-of-sample performance is a better criterion to assess forecasting performance. The next section will introduce this concept.

Section II: Realistic Forecast Evaluation

This section illustrates how to carry out a more realistic forecast evaluation of the nowcasting models presented above, mimicking operational procedures typically adopted in a forecasting unit of a central bank/treasury. We will describe a regularly updated (here once every month) nowcasting model that uses all the available data at the time of the nowcast.

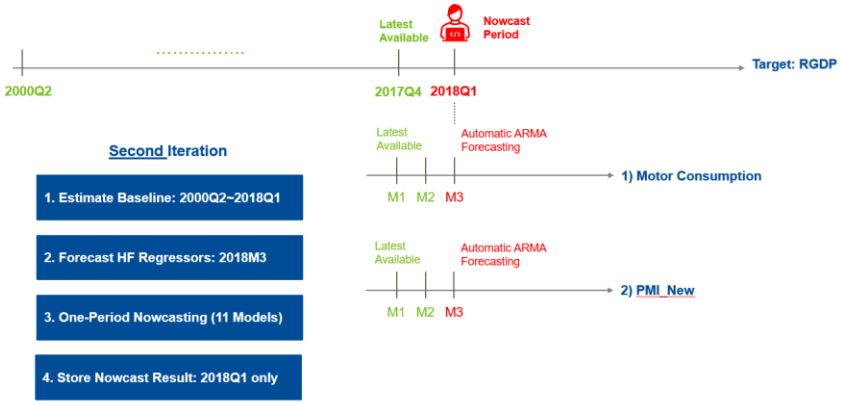
Suppose we are currently at the end of 2018M1 and wish to nowcast the 2018Q1 value of **rgdp**, which is not yet published. Assume that all monthly indicators are available until 2018M1. Therefore, to nowcast the 2018Q1 value of **rgdp**, the first step is to forecast monthly indicators for 2018M2-2018M3 and then use the extended monthly series to nowcast the quarterly **rgdp** value for 2018Q1 (Figure 5).

Figure 5: Realistic Forecast Evaluation (First Month)



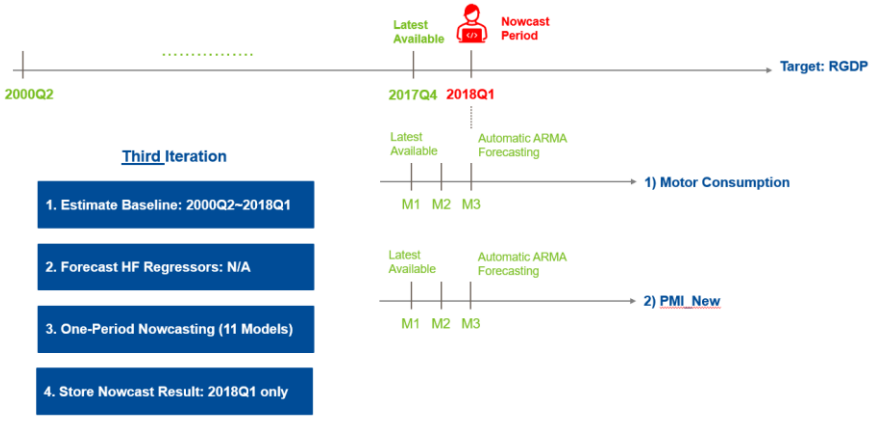
At the end of the next month, 2018M2 data becomes available for all monthly variables. So, we update the monthly data series accordingly. We now need to forecast monthly indicators by only one month (i.e., 2018M3) to update the quarterly **rgdp** nowcast for 2018Q1 (Figure 6).

Figure 6: Realistic Forecast Evaluation (Second Month)



By the end of the quarter, all the required monthly data are available. Thus, we do not need to forecast the monthly indicators to update the 2018Q1 **rgdp** nowcast (Figure 7).

Figure 7: Realistic Forecast Evaluation (Third Month)



Once the official **rgdp** is released (assume at the beginning of next quarter), we will re-estimate the baseline model using the most up-to-date data and repeat the previous steps until the end of the realistic evaluation period.

After several quarters of repeating the above steps, we can evaluate the forecast accuracy of different models based on the usual standard forecast evaluation statistics such as RMSE, MAE, MAPE, etc. This process updates the base forecasting regression at the end of each quarter and uses 1-step ahead nowcasts to evaluate the model's forecasting performance.

The first iteration of the procedure will be demonstrated in detail below, while the complete iterations can be performed by running the "**9_Realistic_Forecast_Evaluation.prg**" program file.

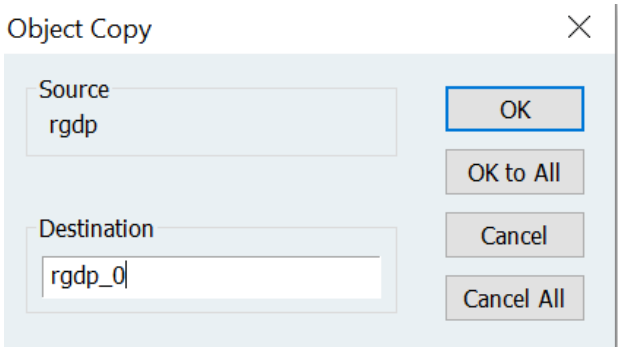
The evaluation sample is 2018Q1 to 2023Q1, and we will start with the nowcast for 2018Q1.

Step 1. Create an RGDP series that reflects the actual data available in 2018M1 (end).

To best emulate actual forecasting/nowcasting practice, assume that **rgdp** is available until one period before the forecast (i.e., 2017Q4 in the first iteration). Mimic this "reality" by creating a new **rgdp** series that contains actual data only up to 2017Q4.

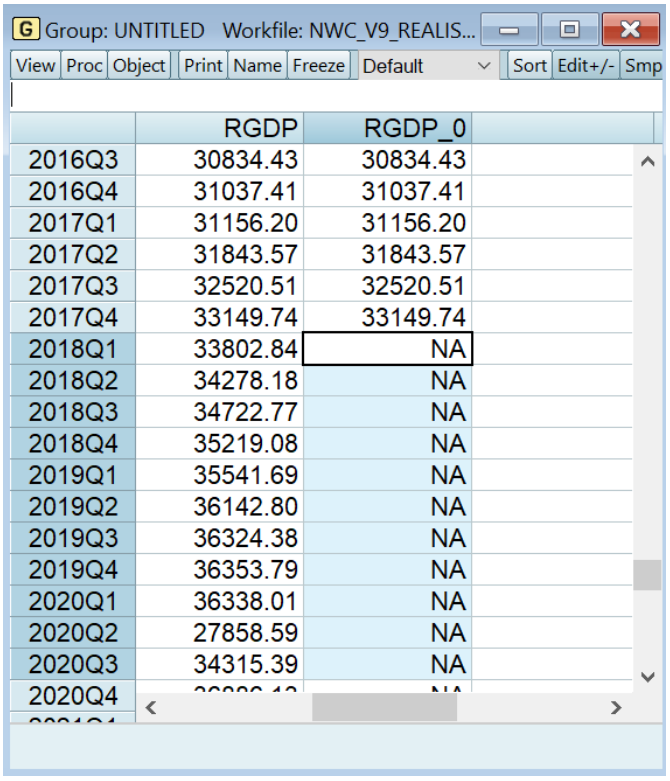
Instruction: click "**rgdp**" series => "copy" => "paste" => destination: "**rgdp_0**" => "**OK**".

Figure 8: copy and paste rgdp series



Then manually remove the actual values of real GDP for 2018Q1-2023Q1 in the **rgdp_0** series.

Figure 9: Removed values of 2018Q1-2023Q1 for rgdp_0



	RGDP	RGDP_0
2016Q3	30834.43	30834.43
2016Q4	31037.41	31037.41
2017Q1	31156.20	31156.20
2017Q2	31843.57	31843.57
2017Q3	32520.51	32520.51
2017Q4	33149.74	33149.74
2018Q1	33802.84	NA
2018Q2	34278.18	NA
2018Q3	34722.77	NA
2018Q4	35219.08	NA
2019Q1	35541.69	NA
2019Q2	36142.80	NA
2019Q3	36324.38	NA
2019Q4	36353.79	NA
2020Q1	36338.01	NA
2020Q2	27858.59	NA
2020Q3	34315.39	NA
2020Q4		

Step 2. Nowcast the High-Frequency Dependent Variables

To simulate realistic nowcasting in practice, we need to use only the data that are available at the time of the nowcasting. We will assume that the latest quarterly data for the RHS variables is published at the time of the forecast, while the monthly data are available up to one month prior to the nowcast target date.[2] This implies forecasting the monthly data for two months (2018M2-

[2] These assumptions can be adjusted based on the country's data release date.

2018M3) by the automatic ARIMA forecasting technique. Doing so ensures non-missing observations for the high-frequency indicators for 2018Q1.

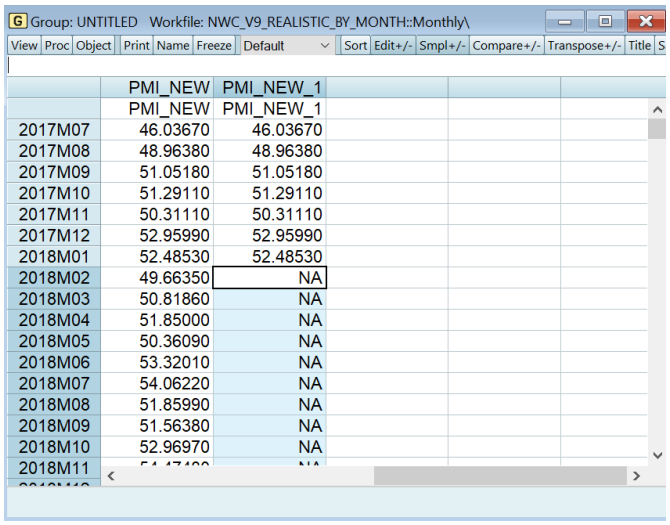
For example, to nowcast **pmi_new** (see "Monthly" pagefile) requires the following three steps:

1. Copy and paste the **pmi_new** series (exactly as **rgdp**). Name the copied series as "**pmi_new _0**".

2. Remove the actual data values in the **pmi_new _0** series from 2018M02 to 2022M12.

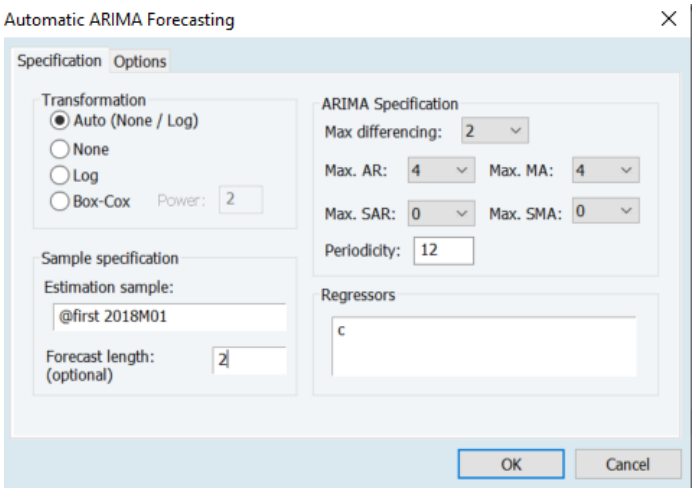
3. Invoke "Proc" => "Automatic ARIMA Forecasting" => Enter the estimation sample and a forecast length of 2 months as shown in Figure 11.

Figure 10: Nowcast for PMI



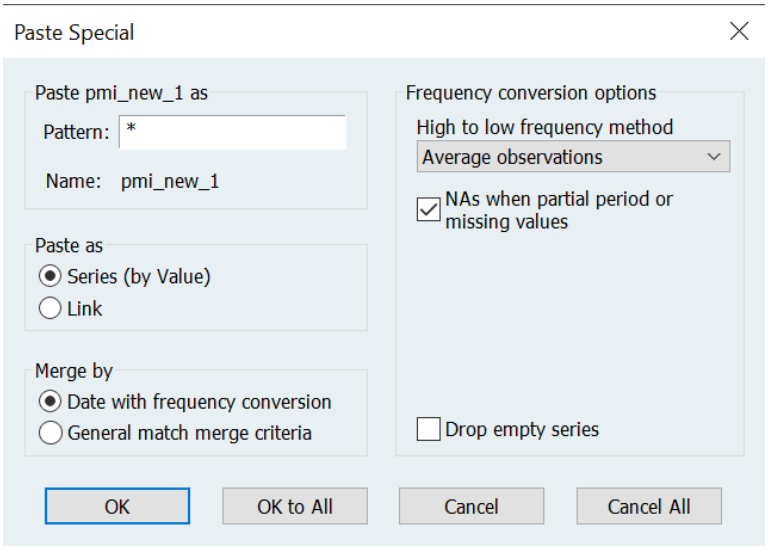
	PMI_NEW	PMI_NEW_1
	PMI_NEW	PMI_NEW_1
2017M07	46.03670	46.03670
2017M08	48.96380	48.96380
2017M09	51.05180	51.05180
2017M10	51.29110	51.29110
2017M11	50.31110	50.31110
2017M12	52.95990	52.95990
2018M01	52.48530	52.48530
2018M02	49.66350	NA
2018M03	50.81860	NA
2018M04	51.85000	NA
2018M05	50.36090	NA
2018M06	53.32010	NA
2018M07	54.06220	NA
2018M08	51.85990	NA
2018M09	51.56380	NA
2018M10	52.96970	NA
2018M11		

Figure 11: Nowcast for PMI



Now that we have **pmi_new** extended out to 2018M03, we can convert it to the quarterly frequency. To do so, copy the monthly (nowcast) series **pmi_new_0_f** and then invoke "past special" in the "Quarterly" pagefile. Select "Sum observations" or "Average observations" in the "High to low frequency method" section (i.e., **pmi_new** has been forecast out to 2018Q1).

Figure 12: Copy and Paste Special for pmi_new_0



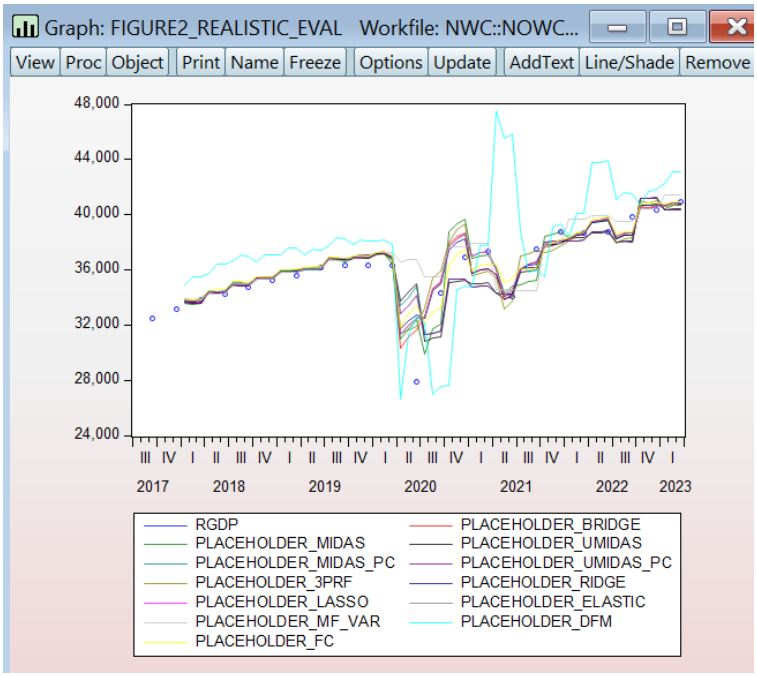
Lastly, repeat the same procedure for the monthly series **consu_motor**.

Step 3. Calculate the Nowcast Models

In step 3, you may now use the eleven nowcast models to nowcast **rgdp** for 2018Q1. Please refer to the specific instructions in the previous section as shown below.

1. **Bridge**: Workshop 4
2. **MIDAS**: Workshop 4
3. **U-MIDAS**: Workshop 4
4. **Principal Components (MIDAS)**: Workshop 5
5. **Principal Components (U- MIDAS)**: Workshop 5
6. **Three Pass Filter Regressions**: Workshop 5
7. **Dynamic Factor Models**: Workshop 6
8. **Mixed-Frequency VAR**: Workshop 7
9. **Machine Learning (Ridge, LASSO, and Elastic Net)**: Workshop 8

Figure 13: Visualization of Nowcasting Results (Realistic Evaluation)[3]



Step 4. Nowcast Models Evaluations

After generating a one-step-ahead nowcast using each of these models, one can then evaluate the models' forecasting performance using FORCEVAL function as previously demonsatrated.

Figure 14: Forecast Evaluation: Forecast/Nowcast Conducted on the First Month of Each Quarter

Evaluation statistics

Forecast	RMSE	MAE	MAPE	SMAPE	Theil U1	Theil U2
BRIDGE_1	1029.809	783.3837	2.219982	2.218942	0.014128	0.366095
MIDAS_1	1563.266	1009.434	2.924420	2.937844	0.021442	0.609746
UMIDAS_1	1641.248	1050.715	3.059823	3.035911	0.022572	0.605619
MIDAS_PC_1	1677.464	1063.325	3.115019	3.069735	0.023046	0.608501
UMIDAS_PC_1	1525.865	1009.429	2.924859	2.907180	0.020981	0.559195
THREE_PRF_1	1073.712	801.8867	2.288851	2.269942	0.014704	0.371403
RIDGE_1	1340.187	867.2214	2.542022	2.499618	0.018370	0.472014
ELASTIC_1	1430.526	901.1109	2.664799	2.601211	0.019583	0.500860
LASSO_1	1430.418	901.0596	2.664643	2.601067	0.019582	0.500823
FC_1	1344.834	939.8704	2.744269	2.700526	0.018417	0.477104
MF_VAR_1	2130.545	969.0268	2.966321	2.783788	0.029052	0.738077
DFM_1	3536.281	2196.093	6.049539	5.917847	0.047795	1.212362

[3] *Figure2_Realistic Eval* after running the program

Figure 15: Forecast Evaluation: Forecast/Nowcast Conducted on the Second Month of Each Quarter

Evaluation statistics

Forecast	RMSE	MAE	MAPE	SMAPE	Theil U1	Theil U2
BRIDGE_2	974.2167	675.7538	1.945749	1.923896	0.013339	0.335683
MIDAS_2	1396.380	843.1535	2.478614	2.446799	0.019116	0.516570
UMIDAS_2	1691.280	1050.531	3.085457	3.038158	0.023248	0.615555
MIDAS_PC_2	1770.986	1067.904	3.152623	3.085271	0.024319	0.638234
UMIDAS_PC_2	1605.799	1021.105	2.982485	2.946572	0.022069	0.583917
THREE_PRF_2	1117.065	780.9195	2.252681	2.221258	0.015276	0.387494
RIDGE_2	1291.175	728.7779	2.169503	2.104760	0.017665	0.443842
ELASTIC_2	1395.006	742.5789	2.235184	2.149043	0.019060	0.480024
LASSO_2	1394.918	742.5639	2.235140	2.149009	0.019059	0.479994
FC_2	1293.688	802.0516	2.384109	2.325822	0.017687	0.456092
MF_VAR_2	2130.545	969.0268	2.966321	2.783788	0.029052	0.738077
DFM_2	3535.056	2299.651	6.434605	6.255084	0.047493	1.285204

Figure 16: Forecast Evaluation: Forecast/Nowcast Conducted on the Third Month of Each Quarter

Evaluation statistics

Forecast	RMSE	MAE	MAPE	SMAPE	Theil U1	Theil U2
BRIDGE_3	1044.619	678.2968	1.973451	1.937896	0.014282	0.364241
MIDAS_3	1433.933	826.5685	2.454167	2.400210	0.019608	0.518452
UMIDAS_3	1741.772	1024.135	3.030600	2.966029	0.023932	0.629249
MIDAS_PC_3	1855.760	1057.561	3.151310	3.056114	0.025469	0.661504
UMIDAS_PC_3	1664.124	1010.159	2.975075	2.917212	0.022857	0.597906
THREE_PRF_3	1141.596	731.4903	2.122304	2.076841	0.015589	0.404118
RIDGE_3	1350.292	746.9480	2.235676	2.157620	0.018448	0.466375
ELASTIC_3	1474.043	793.6711	2.397740	2.295823	0.020108	0.509853
LASSO_3	1473.956	793.6033	2.397543	2.295636	0.020107	0.509823
FC_3	1341.201	803.7955	2.403960	2.330877	0.018320	0.466833
MF_VAR_3	2130.545	969.0268	2.966321	2.783788	0.029052	0.738077
DFM_3	3535.056	2299.651	6.434605	6.255084	0.047493	1.285204

The Realistic Forecast Evaluation (Out-of-Sample Evaluation) suggests **BRIDGE** is the best nowcasting model. Figures 17 and 18 plot real-time out-of-sample RMSE for each month of a quarter.

Figure 17: Out-of-Sample RMSE Evolution for Best-Performing Model

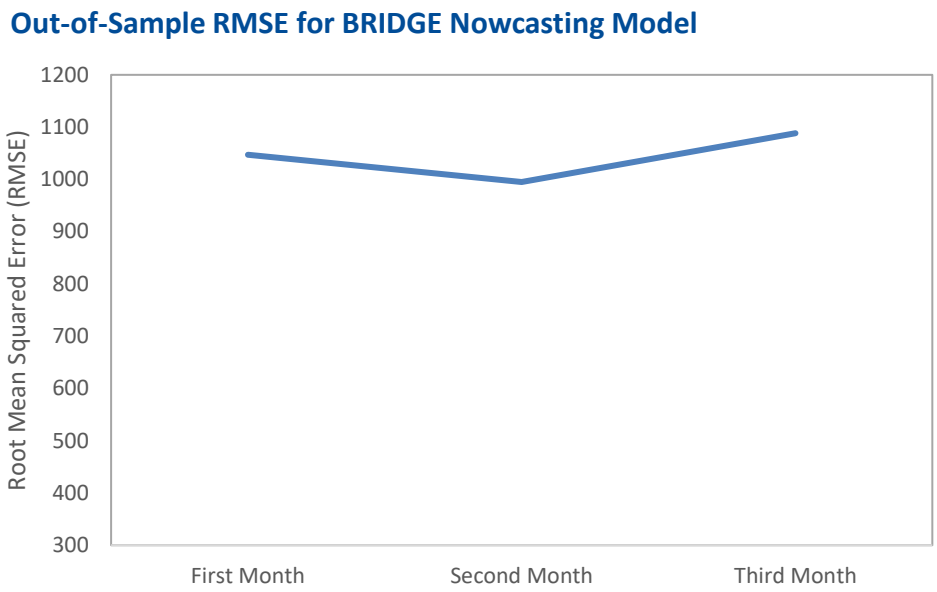
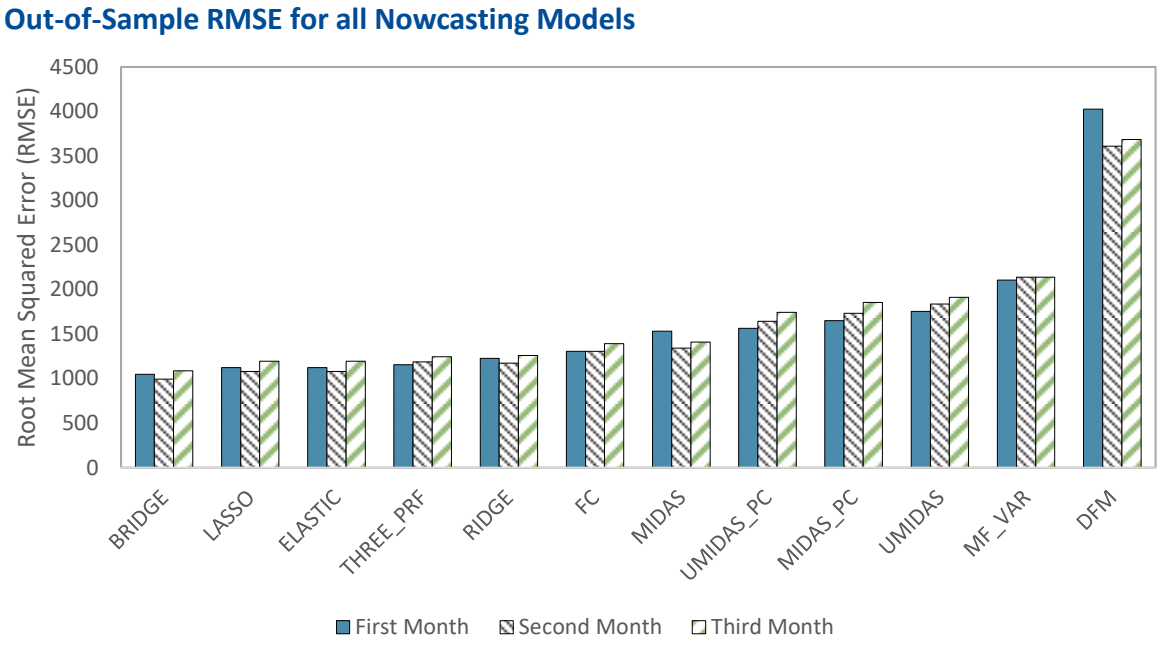


Figure 18: Out-of-Sample RMSE Evolution for All Nowcasting Models (Rank by RMSE value)



Appendix A: EViews Command Line Approach

As demonstrated above, EViews offers an intuitive, menu-based approach to generating forecasts/nowcasts. The same results can also be generated using EViews' programming language and estimation/forecasting objects.

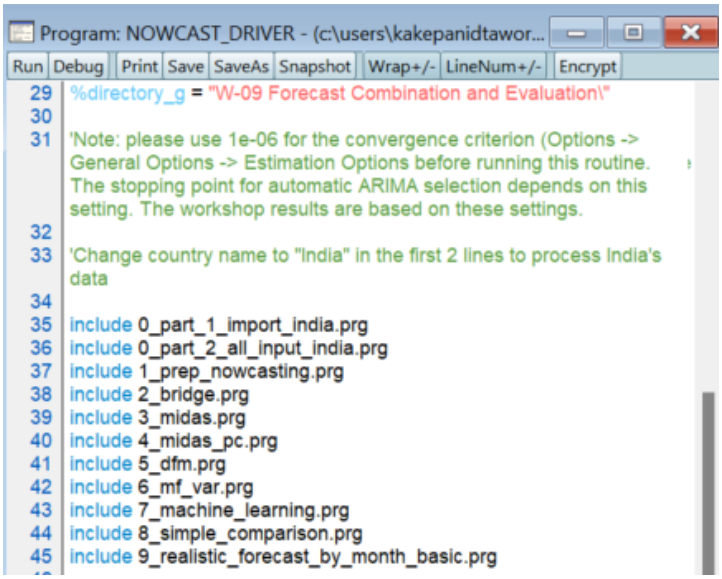
The EViews codes to replicate the workshop results presented above are in the following files:

- nowcast_driver (driver program)
- 0_part_1_import
- 0_part_2_all_input
- 1_prep_nowcasting
- 2_bridge
- 3_midas
- 4_midas_pc
- 5_dfm
- 6_mf_var
- 7_machine_learning
- 8_simple_comparison
- 9_realistic_forecast_by_month_basic

The current working directory needs to be updated to run the program. The driver program loads and executes the commands in all programs.

Click on "Run" to execute the commands in the eleven files and obtain the results cited in this workshop.

Figure 19: Executing the Nowcasting Driver Program for India



A basic program for the simple evaluation part is provided: “8_SNE_BASIC.prg”



If you wish to combine several models given different realistic evaluation performance, you could follow codes below:

For example, if you wish to combine Bridge and Ridge models:

```
Pageselect quarterly_re

Genr fc_special_1 = (bridge_1 + ridge_1)/2

Genr fc_special_2 = (bridge_2 + ridge_2)/2

Genr fc_special_3 = (bridge_3 + ridge_3)/2

' loop from 1 to 3 (from split sampling)

for !i = 1 to 3

    ' set up sample

    smpl {nowcast_eval_start} {estimation_end}

    ' create forecast evaluation

    freeze(forceval_special_{!i}) raw_rgdp.forceval Bridge_!i Midas_!i UMIDAS_!i
midas_pc_!i  umidas_pc_!i  three_prf_!i ridge_!i elastic_!i lasso_!i FC_!i  MF_VAR_!i DFM_!i
FC_SPECIAL_!i

next
```

Appendix B: How to Run Realistic Evaluation Program

This appendix outlines the steps required to configure the *9_realistic_forecast_by_month_basic_india.prg* program so that you can run **realistic forecast evaluations independently**.

1. Setting Up for the Program File

For the program to run successfully from start to finish, please ensure that your EViews workfile satisfies **all** of the following requirements:

1. The target variable name must be written **exactly** as "RGDP" (case-sensitive).
2. The workfile you are using must be named **nwc_8_sc.wf**.
3. Your workfile must contain both a **monthly** page and an **SC (Simple Comparison that contains all your outputs under target frequency)** page.
4. All nowcasting models must already be created in the workfile and named **exactly as specified during the workshop**.
5. The file **threepassfilter.prg** must be saved in the **same working directory** as both your program file and workfile.

Once these requirements are met, you may open the file *9_realistic_forecast_by_month_basic_india.prg* and proceed to configure the necessary inputs for the realistic evaluation.

2. Required Input Sections in the Program File

The program requires **three input sections**, all located at the beginning of the program file (approximately lines 36–90). These sections must be reviewed and customized based on your model and data setup.

Input #1: Baseline Equation Specification

This section defines the baseline regression used for nowcasting. You should specify:

- The dependent (target) variable
- Independent variables
- Crisis dummy variables (if applicable)
- Any required data transformations

```
INPUT NEEDED #1. Model Specification
============================================================
1) Nowcasting target variable, with its transformation
%nwc_target_variable = "DLOG(RGDP)"

2) Lags of the dependent variable to include in the regressions
%lagged_dependent_variables = "DLOG(RGDP(-3)) DLOG(RGDP(-2)) DLOG(RGDP(-1))"

3) Exogenous variables at the target (quarterly) frequency (leave "" if none)
%exogenous_variables_target_frequency = "DLOG(RGDP_US(-1)) DLOG(RGDP_CN(-1))"

4) Dummy variables to include (leave "" if none)
%included_dummy_variables = "CRISIS"

5) High-frequency (monthly) indicators, in levels
%high_frequency_variables = "CONSU_MOTOR PMI_NEW"

6) Transformation of each indicator above, listed in the same order
%high_frequency_transformations = "DLOG(CONSU_MOTOR) D(PMI_NEW)"

7) Monthly-to-quarterly conversion for each indicator, listed in the same order
sn = sum aggregate; an = average aggregate
%high_frequency_conversion_methods = "sn an"

8) Define variables included in the Dynamic Factor Model
%variables_in_dfm = "RGDP CONSU_MOTOR PMI_NEW"
```

Input #2: Evaluation and Estimation Periods

This section defines the start and end dates for:

- The estimation sample
- The realistic forecast evaluation period

```
' ============================================================
' INPUT NEEDED #2. Sample Periods (quarterly dates only)
' ============================================================
' 1) Estimation sample
%estimation_start_quarterly = "2000Q1"
%estimation_end_quarterly = "2023Q1"

' 2) Nowcast evaluation period
%nowcast_eval_start_quarterly = "2018Q1"
%nowcast_eval_end_quarterly = "2023Q1"

' 3) Nowcast target period
%nowcast_target_end_quarterly = "2023Q2"
```

Input #3: Input and Output pages

This section specifies input and output pages and workfiles.

```
' ============================================================
' INPUT NEEDED #3. Page Holding the Nowcasting Models
' ============================================================
' 1) Workfile produced by the previous program
%input_workfile = "nwc_8_sc.wf1"

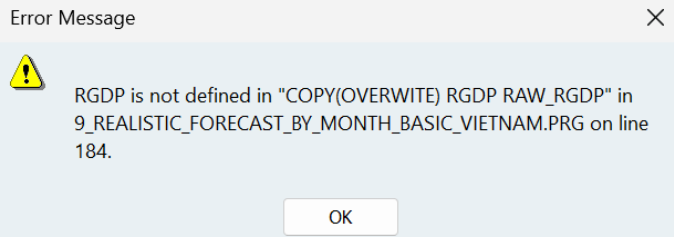
' 2) Name of the page in the workfile where the Bridge, MIDAS and U-MIDAS
' equations and their in-sample forecasts were stored by the earlier programs
%nowcast_models_page = "MIDAS"

' 3) ' Name under which the workfile is saved once the charts are built.
' Set it to %input_workfile if you prefer to overwrite the input workfile.
%output_workfile = "nwc_9_realistic.wf1"
```

3. Common Errors and How to Resolve Them

Below are some common errors you may encounter when running the program, along with suggested solutions.

Error #1: RGDP not defined. This error typically occurs when the target variable in your workfile is named something other than "RGDP", causing the program to fail when referencing it.

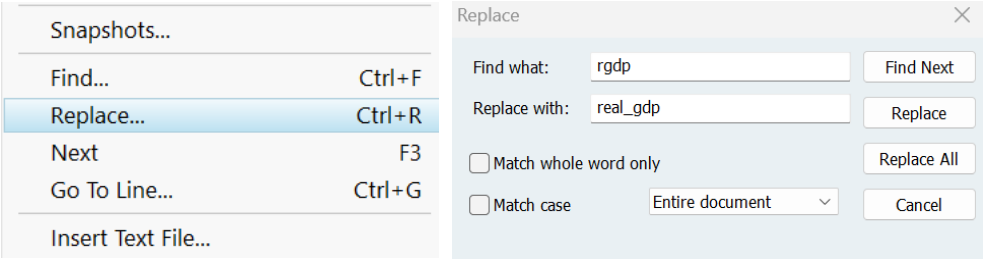


Potential solution #1. The simplest fix is to replace all instances of "RGDP" in the program with the actual name of your target variable.

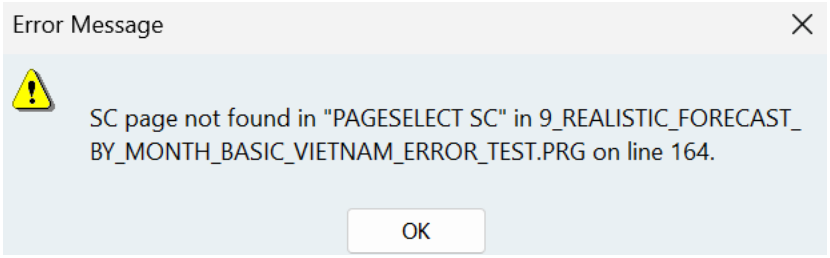
To do this:

- Right-click anywhere in the program editor
- Select **Replace**
- Replace "RGDP" with your target variable name (e.g., "real_gdp")
- Click **Replace All**

This will allow the program to run correctly using your variable naming convention.



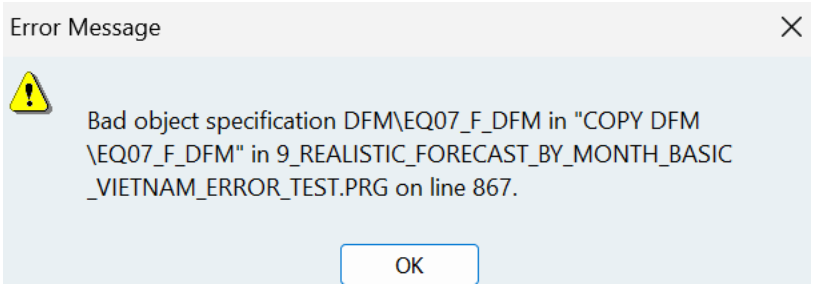
Error #2: SC page not found. This error occurs when the program attempts to access the "SC" page, but no such page exists in your workfile.



Potential solution #2. In line 164 of the program, replace "SC" with the name of the page that stores your quarterly outputs. For example, if your quarterly output page is named "Simple_compare", change “SC” to “Simple_compare” in line 164.



Error #3: Bad object specification. This error usually arises when some nowcasting model outputs are missing, but the program is still attempting to copy all equations into the summary page.



Potential solution #3. Comment out the lines that generate the error by adding an apostrophe (') at the beginning of the affected lines. This prevents EViews from executing those commands.

```
867 ' copy dfm\eq07_f_dfm
```

4. Review Outputs

After the program runs successfully, a new EViews workfile named **NWC_9_Realistic_Evaluation.wf1** will be created automatically. All major outputs, including model estimations, in-sample evaluations, and out-of-sample evaluations, are stored in the **NOWCAST_RESULT** page.

The main outputs for the realistic evaluation include the following:

- **Figure2_realistic_eval**
  This figure illustrates how closely each nowcasting model tracks actual GDP across different nowcast rounds within each year.

- **Table2_realistic_eval_1 to Table4_realistic_eval_3**
  These tables summarize out-of-sample evaluation metrics for nowcasts produced at the end of the first, second, and third month of each quarter, respectively.

If you encounter additional errors that are not covered here or cannot be resolved using the suggested solutions, please feel free to contact: Jing Xie (jxie5@outlook.com).